

Diagrama Base Datos

Diagrama de Base de Datos

Proyecto: QHere - Control de Asistencia Escolar

[Diagrama Mermaid]

erDiagram

schools ||--o{ profiles : asigna

schools ||--o{ grade_sections : contiene

schools ||--o{ schedules : define

schools ||--o{ school_calendar : planifica

schools ||--o{ students : registra

schools ||--o{ authorized_devices : autoriza

schools ||--o{ notification_queue : genera

profiles ||--o{ parents : vincula

profiles ||--o{ student_teachers : imparte

profiles ||--o{ attendance : registra

profiles ||--o{ excuses : revisa

profiles ||--o{ audit_log : ejecuta

grade_sections ||--o{ students : agrupa

students ||--o{ parents : pertenece

students ||--o{ student_teachers : recibe

students ||--o{ attendance : genera

students ||--o{ excuses : justifica

attendance ||--o{ excuses : relaciona

attendance ||--o{ attendance_geo_events : ubica

schools {

uuid id PK

text nombre

text direccion

text telefono

text email

text director

date academic_period_start

date academic_period_end

numeric latitude

numeric longitude

numeric allowed_radius_m

boolean configurado

}

profiles {

uuid id PK

uuid school_id FK

text full_name

text email

text role

text approval_status

text phone

text[] permisos

uuid[] secciones_ids

}

grade_sections {

uuid id PK

uuid school_id FK

text grado

text seccion

text turno

boolean special_schedule_enabled

time hora_entrada_especial

time hora_salida_especial

time hora_limite_tardanza_especial

}

schedules {

uuid id PK

uuid school_id FK

text turno

time hora_entrada

time hora_salida

time hora_limite_tardanza

}

school_calendar {

uuid id PK

uuid school_id FK

date fecha

text tipo

text descripcion

}

students {

uuid id PK

uuid school_id FK

uuid grade_section_id FK

text nombre

text matricula

text full_name

text enrollment_code

uuid parent_id FK

text qr_token

boolean activo

}

attendance {

uuid id PK

uuid student_id FK

uuid teacher_id FK

date fecha

time hora_entrada

time hora_salida

text estado

text dispositivo

integer minutos_tarde

}

excuses {

uuid id PK

uuid student_id FK

uuid parent_id FK

uuid attendance_id FK

uuid teacher_id FK

date absence_date

text reason

text attachment_url

text status

}

Notas tecnicas

- `profiles` se relaciona con Supabase Auth por medio de `auth.users(id)`.
- `students.qr\_token` identifica el QR unico de cada estudiante.
- `attendance` registra entrada, salida, tardanza, dispositivo y relacion con docente.
- `audit\_log` conserva trazabilidad de acciones sensibles.
- `authorized\_devices` y `attendance\_geo\_events` cubren control de dispositivo y ubicacion.